CLASS SESSIONS
Wednesdays, 1:00 – 3:50 PM
Hammer 303

INSTRUCTOR
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TEACHING ASSISTANT
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Office Hours: Mondays, 12:00-1:00 PM (via Zoom or emails)

COURSE DESCRIPTION
This is an applied statistical methods course. The course will review the fundamental features of complex sample designs, including simple random sampling, probability-proportional-to-size sampling, stratified sampling, cluster sampling, and multistage sampling. Students will learn how to construct survey weights, including developing design weights, classifying response status codes, adjusting design weights for unknown eligibility and nonresponse, and calibrating the adjusted weights to the known populations using poststratification and raking. Using national health surveys as examples, the course will introduce and demonstrate the application of survey statistical methods in generating descriptive statistics and fitting regression models.

PREREQUISITES
P8104: Probability
P8131: Biostatistical Methods II

CEPH DEPARTMENTAL DEGREE PROGRAM COMPETENCIES
This course is designed to help students attain mastery of the following degree competencies. Student achievement of these competencies will be measured through performance on the corresponding assessments.

CEPH Competency	Primary Assessment(s)
DrPH: Explain the use and limitations of surveillance systems and national surveys in assessing, monitoring, and evaluating policies and programs and to address a population's health.	In the presentation project, students are assessed using survey data from NHANES. Students must explain components of the surveys, including sample design, survey weights, and data collection and analysis. Because this is a group project, DrPH students will be assessed individually through both a peer and self-evaluation that includes a question on how students work contributed to mastery of the competency.

MS: Apply principles of statistical design to answer public health research questions.	In the presentation project, students are asked to present the sampling document of a national survey, which tests students on the learned concepts of survey design.

COURSE LEARNING OBJECTIVES

Students who successfully complete this course will be able to:

- understand the main techniques used in sampling practice: simple random sampling, systematic selection, probability proportional to size sampling, stratification, cluster sampling, multistage sampling
- construct weights for complex surveys
- estimate variance using replication methods, including the Jackknife, balanced half-sample replication, and the Bootstrap
- analyze health surveys, including graphic tools, univariate and bivariate analyses, and regression models
- implement multiple imputation for missing data in surveys

TEXBOOKS

Recommended textbooks:

- Kish, L. (1965). *Survey Sampling*. Wiley.
- Korn, EL & Graubard, BI (1999). *Analysis of Health Surveys*. John Wiley & Sons. (Columbia ebook)
- Heeringa, SG, West, BT and Berglund, PA (2017). *Applied Survey Data Analysis. Second Edition*. Chapman & Hall/CRC, Boca Raton, FL. (Columbia e-book)
- Valliant, R, Dever, JA and Kreuter, F. (2013). *Practical Tools for Designing and Weighting Survey Samples*. Springer, Ne York. (Columbia e-book)

ASSESSMENT AND GRADING POLICY

Student grades will be based on:

Homework.....30%
Quizzes.....35%
Survey document presentation.....10%
Final project.....20%
Class participation 5%

Grading

A+ Reserved for highly exceptional achievement.
A Excellent. Outstanding achievement.
A- Excellent work, close to outstanding.
B+ Very good. Solid achievement expected of most graduate students.
B Good. Acceptable achievement.
B- Acceptable achievement, but below what is generally expected of graduate students.
C+ Fair achievement, above minimally acceptable level.
C Fair achievement, but only minimally acceptable.
F Failure. Course usually may not be repeated unless it is a required course.

COURSE REQUIREMENTS

The course will consist of lectures, homework assignments, quizzes, survey document presentation, and a final project. Academic integrity is a core value at Mailman. For this reason, the use of generative artificial intelligence (AI) sites, (for example, but not only, Chat GPT) to complete an assignment, quiz, or project **is not permitted unless the course instructor has provided clear written instruction about the use of generative AI.**

- In each homework, students will be provided with a set of questions, which will require either hand calculations, statistical analyses using R, or programming in R. The homework should be submitted via the Canvas. **No late homework will be accepted.** Students are encouraged to work together on homework, although only in the spirit of learning. There should be no copying of work. The homework with the lowest score will not be counted toward the final grade.
- Five short quizzes will be given at the beginning of classes.
- Students will be assigned with sampling documents of national surveys and do a presentation on the assigned sampling documents.
- Students will work on a final project, using the statistical methods learned in the class to analyze a national survey dataset and write a report.
- Class participation will be evaluated via class attendance and in-class discussion. **Students are required to email the TA and copy the instructor before the class starts if they cannot attend the class due to medical reasons or family emergency.**

COURSE STRUCTURE

The class sessions will be primarily lectures supplemented with guest seminars given by researchers working on survey methods and practice.

MAILMAN SCHOOL POLICIES AND EXPECTATIONS

Students and faculty have a shared commitment to the School's mission, values and oath. mailman.columbia.edu/about/mission-history

Academic Integrity

Students are required to adhere to the Mailman School Conduct and Community Standards, which includes the Code of Academic Integrity. Columbia Mailman and Columbia University take academic integrity very seriously. This instructor and course are no different. Should any student be suspected of an academic integrity violation, there will be a report submitted to the Center for Student Success & intervention/Student Conduct. After these offices conduct their process, if a student is found responsible for violating an academic integrity policy (see Standards & Discipline/Academic Violations and Student Honor Code & Professional Guidelines), they will be assigned a grade penalty, with a possible outcome being a 0% on the assignment. *[note to faculty: please update this according to your course specific policy]*

Please review the university, school, and course policies, as you are responsible for behaving according to the outlined expectations.

Personal Support

Students sometimes experience life challenges that require additional support and connection to resources. If you are experiencing difficult circumstances, please reach out for help and support. Student Support Services in the Office of Student Affairs is poised to connect with students,

provide resource referrals, and provide ongoing, non-clinical support. They are a good place to start if you do not know where to turn.

If you would like to make a referral to student support, whether for yourself or someone else, the best way to do so is to fill out the form linked below, and student support will reach out directly. You may also email Meurcie Zignoli at mz3047@cumc.columbia.edu to connect to Student Support Services. Please note that this form is separate from other reporting structures in place for code of conduct violations or Title VI/ Title IX concerns. This form is also not an emergency response mechanism. In case of an emergency, please contact Public Safety at 212-305-7972: Mailman Student Support Team Referral Form (maxient.com)

Disability Access
In order to receive disability-related academic accommodations, students must first be registered with the Office of Disability Services (ODS). Students who have or think they may have a disability are invited to contact ODS for a confidential discussion at 212.854.2388 (V) 212.854.2378 (TTY), or by email at disability@columbia.edu. If you have already registered with ODS, please speak to your instructor to ensure that they have been notified of your recommended accommodations by Meredith Ryer (mr4075@cumc.columbia.edu), Assistant Director of Student Support and Mailman's liaison to the Office of Disability Services.

Bias Incidents
Our community at Columbia University's Mailman School of Public Health is committed to creating an inclusive working, learning, and living environment where all are respected. The occurrence of bias related incidents, involving conduct, speech, or expressions reflecting prejudice are an opportunity for learning and growing as a community.

As part of our efforts to create as inclusive a community as possible, when bias incidents occur at Columbia, we provide an opportunity for those involved to engage in education, advocacy and conversation. In this way, we work to address the incident and minimize the potential for future occurrences. Our community's tools to address bias include a reporting process and the Bias Incident Resource Team, plus resources within schools and various offices. You can access information about the Bias Reporting Process and FAQs here.

Why Reporting Matters and How to File a Report
Our priority is ensuring that Columbia University is a safe community and workplace where we can learn, live, work and express ourselves. As members of the community, we have a shared responsibility to uphold these standards and report behavior that violates these standards. The reporting options provide Columbia University community members an opportunity to share important information directly with appropriate offices. If you or a member of the community needs support please take the time to complete a report so we may provide support, care, and accountability: https://universitylife.columbia.edu/report

COURSE SCHEDULE

Session 1 – Introduction to survey sampling	
09/03	Learning Objectives: ▪ Go through the syllabus ▪ Provide an overview of health surveys ▪ Define sampling frame and frame problems ▪ Learn what is probability sampling ▪ Introduce sampling methods: simple random sampling, stratified sampling Assignment: HW #1 assigned

Session 2 – Sampling methods	

09/10	Learning Objectives:
	▪ Learn sampling methods
	  ○ Systematic sampling
	  ○ Probability proportional to size (PPS) sampling
	  ○ Systematic PPS sampling
	  ○ Cluster sampling

Session 3 – More sampling methods	

09/17	Learning Objectives:
	▪ Quiz 1
	▪ Learn sampling methods
	  ○ Stratified PPS sampling of clusters
	  ○ multistage area probability sample design
	▪ Watch lab recordings
	Assignment:
	HW #2 assigned

Session 4 – Sample size determination	
9/24	Learning Objectives: ▪ Sample size determination for SRS & stratified sample & 2-tage sampling

Session 5 – Construct survey weights (I)	
10/01	Learning Objectives: ▪ Quiz 2 ▪ Outcome rates ▪ Weight components ▪ Base weight ▪ Adjustment for unknow eligibility Assignment: HW #3 assigned

Session 6 – NO CLASS
10/8

Session 7 – Construct survey weights (II)	
10/15	Learning Objectives: ▪ Quiz 3 ▪ Adjustment for nonresponse Assignment: HW #4 assigned

Session 8 – Construct survey weights (III) & Variance estimation	
10/22	Learning Objectives: ▪ Poststratification and raking ▪ Linearization method ▪ Replication variance estimation methods: ○ Jackknife

Session 9 – Replicate weights	

10/29	Learning Objectives:
	▪ Quiz 4
	▪ Replication variance estimation methods:
	  ○ Balanced half-sample replication
	  ○ Bootstrap
	▪ Replicate weights with R
	Assignment:
	HW #5 assigned
	Survey document presentation project assigned

Session 10 – Estimation of descriptive statistics	
11/5	Learning Objectives: ▪ Preparation for complex survey data analysis ▪ Graphical tools for survey data ▪ Estimation of percentiles ▪ Domain estimation

Session 11 – Analysis of categorical data of complex surveys	
11/12	Learning Objectives: ▪ Quiz 5 ▪ Estimation of ratios ▪ Linear regression for survey data

Session 12 – Regression of complex survey data

11/19	Learning Objectives: ▪ Univariate categorical data analysis ▪ Bivariate categorical data analysis ▪ Logistic regression for survey data Assignment: HW #6 assigned Survey document presentation is due

Session 13 – Missing data and multiple imputation	
12/03	Learning Objectives: ▪ Define missing data mechanisms ▪ Implement multiple imputation ▪ Statistical analysis using multiply imputed data Assignment: HW #7 assigned Final project assigned

Session 14 – Guest lecture/ Final project Q & A	
12/10	Learning Objectives: ▪ Guest lectures/ Q & A for final projects ▪ Watch recording: multiple imputation